love.c

```c
#include<stdio.h>
#include<math.h>


int main(){
    /*

    // first program
    printf("hello world !");

    */


    /*

    // sum of two numbers
    int a, b;
    printf("enter number a : ");
    scanf("%d", &a);
    printf("enter number b : ");
    scanf("%d", &b);
    int sum = a + b;
    printf("sum of a and b is : %d\n", sum);

    */


    /*

    // area of square
    float side, area;
    printf("enter side of square : ");
    scanf("%f", &side);
    area = side * side;
    printf("area of square is : %.2f\n", area);

    */


    /*

    // area of circle
    const float pi = 3.14;
    float radius, area;
    printf("enter radius of circle : ");
    scanf("%f", &radius);
    area = pi * radius * radius;
    printf("area of circle is : %.2f\n", area);
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    */


    /*

    // perimeter of rectangle
    float length, breadth, peri;
    printf("enter length and breadth of perimeter : \n");
    scanf("%f %f",&length, &breadth);
    peri = 2 * ( length + breadth );
    printf("perimeter of rectangle is : %.2f\n", peri);

    */


    /*

    // cube of number n
    printf("enter number n : ");
    int n, cube;
    scanf("%d", &n);
    cube = n*n*n;
    printf("cube of number n is : %d\n", cube);

    */


    /*

    // simple interest
    int p, t;
    float r, si;
    printf("enter principle amount : ");
    scanf("%d", &p);
    printf("enter time : ");
    scanf("%d", &t);
    printf("enter rate of interest : ");
    scanf("%f", &r);
    si = (p * r * t) / 100;
    printf("simple interest is : %.2f\n", si);

    */


    /*

    // gross salary
    float bs, gs;

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    printf("enter basic salary : ");
    scanf("%f", &bs);

    float da = 0.4 * bs;
    float hra = 0.2 * bs;

    printf("basic salary is : %.2f\n", bs);
    printf("dearness allowance is : %.2f\n", da);
    printf("house rent allowance is : %.2f\n", hra);

    gs = da + hra + bs;
    printf("gross salary is : %.2f\n", gs);

    */


    /*

    // conversion of distance
    float km, m, cm, inch, ft;

    printf("Enter distance in kilometers: ");
    scanf("%f", &km);

    m = km * 1000;
    cm = m * 100;
    inch = cm / 2.54;
    ft = inch / 12;

    printf("Distance in meters = %f\n", m);
    printf("Distance in centimeters = %f\n", cm);
    printf("Distance in inches = %f\n", inch);
    printf("Distance in feet = %f\n", ft);

    */


    /*
    // calculation of total marks and percentage marks
    int m1, m2, m3, m4, m5, total;
    float percentage;

    printf("Enter marks of 5 subjects: ");
    scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);

    total = m1 + m2 + m3 + m4 + m5;
    percentage = total / 5.0;   // important: 5.0 for float division

    printf("Aggregate Marks = %d\n", total);
    printf("Percentage = %.2f\n", percentage);
```

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    */


    /*
    // Convert temperature from Fahrenheit → Centigrade
    float f, c;

    printf("Enter temperature in Fahrenheit: ");
    scanf("%f", &f);

    c = (5.0 / 9.0) * (f - 32);
    printf("Temperature in Centigrade = %.2f\n", c);
    */


    /*
    // find Area of rectangle, Perimeter of rectangle, Area of circle, Circumference of circle
    float l, b, r;
    float rect_area, rect_peri;
    float circle_area, circle_circ;

    printf("Enter length and breadth of rectangle: ");
    scanf("%f %f", &l, &b);

    printf("Enter radius of circle: ");
    scanf("%f", &r);

    // Rectangle
    rect_area = l * b;
    rect_peri = 2 * (l + b);

    // Circle
    circle_area = 3.14 * r * r;
    circle_circ = 2 * 3.14 * r;

    printf("\nRectangle Area = %.2f\n", rect_area);
    printf("Rectangle Perimeter = %.2f\n", rect_peri);

    printf("\nCircle Area = %.2f\n", circle_area);
    printf("Circle Circumference = %.2f\n", circle_circ);
    */


    /*
    // calculate and print paper sizes A0, A1, A2,..., A8.
    int l = 1189, b = 841;

    printf("A0 = %d x %d\n", l, b);
    l /= 2;
```

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    printf("A1 = %d x %d\n", b, l);
    b/=2;
    printf("A2 = %d x %d\n", l, b);
    l /= 2;
    printf("A3 = %d x %d\n", b, l);
    b/=2;
    printf("A4 = %d x %d\n", l, b);
    l/=2;
    printf("A5 = %d x %d\n", b, l);
    b/=2;
    printf("A6 = %d x %d\n", l, b);
    l/=2;
    printf("A7 = %d x %d\n", b, l);
    b/=2;
    printf("A8 = %d x %d\n", l, b);
    */


    /*
    // average of 3 numbers
    float a, b, c, avg;
    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);
    avg = (a + b + c) / 3;
    printf("Average = %.2f\n", avg);
    */


    /*
    // Check if a character is a digit (0–9) or not
    char ch;
    int result;
    printf("Enter a character: ");
    scanf(" %c", &ch);
    result = (ch >= '0') * (ch <= '9');
    // result = ch >= '0' || ch <= '9';
    printf("Result = %d\n", result);
    */


    /*
    // check smallest of two numbers
    int a, b;
    printf("enter number a : ");
    scanf("%d", &a);
    printf("enter number b : ");
    scanf("%d", &b);
    printf("0 = a is smallest & 1 = b is smallest ---> %d\n", a > b);
    */

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    /*
    // area of triangle using side(a, b, c)
    float a, b, c, sp, area;
    printf("Enter three sides of triangle: ");
    scanf("%f %f %f", &a, &b, &c);
    sp = (a + b + c) / 2;
    area = sqrt(sp * (sp - a) * (sp - b) * (sp - c));
    printf("Area of triangle = %.2f\n", area);
    */


    /*
    // Given: A 5-digit number, Task: Reverse the number
    int n, d1, d2, d3, d4, d5;
    long int rev;

    printf("Enter a 5-digit number: ");
    scanf("%d", &n);

    d5 = n % 10;
    n = n / 10;

    d4 = n % 10;
    n = n / 10;

    d3 = n % 10;
    n = n / 10;

    d2 = n % 10;
    n = n / 10;

    d1 = n % 10;

    rev = d5*10000 + d4*1000 + d3*100 + d2*10 + d1;

    printf("Reversed number = %ld\n", rev);
    */


    /*
    // given --> An amount (₹N), task --> Find minimum number of notes using denominations: 100, 50, 10, 5, 2, 1
    int amount;
    int n100, n50, n10, n5, n2, n1, total;

    printf("Enter amount: ");
    scanf("%d", &amount);

    n100 = amount / 100;
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    amount = amount % 100;

    n50 = amount / 50;
    amount = amount % 50;

    n10 = amount / 10;
    amount = amount % 10;

    n5 = amount / 5;
    amount = amount % 5;

    n2 = amount / 2;
    amount = amount % 2;

    n1 = amount;

    total = n100 + n50 + n10 + n5 + n2 + n1;

    printf("100 notes = %d\n", n100);
    printf("50 notes = %d\n", n50);
    printf("10 notes = %d\n", n10);
    printf("5 notes = %d\n", n5);
    printf("2 notes = %d\n", n2);
    printf("1 notes = %d\n", n1);

    printf("Total notes = %d\n", total);
    */


    /*
    // Input a 5-digit number, Find sum of its digits
    int n, d1, d2, d3, d4, d5, sum;

    printf("Enter a 5-digit number: ");
    scanf("%d", &n);

    d1 = n % 10;
    n = n / 10;

    d2 = n % 10;
    n = n / 10;

    d3 = n % 10;
    n = n / 10;

    d4 = n % 10;
    n = n / 10;

    d5 = n % 10;

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    sum = d1 + d2 + d3 + d4 + d5;

    printf("Sum of digits = %d\n", sum);
    */



    /*
    // Input: angle,  Output: all trigonometric ratios
    float angle, rad;

    printf("Enter angle in degrees: ");
    scanf("%f", &angle);

    // convert to radians
    rad = angle * 3.14 / 180;

    printf("sin = %.2f\n", sin(rad));
    printf("cos = %.2f\n", cos(rad));
    printf("tan = %.2f\n", tan(rad));

    printf("cosec = %.2f\n", 1/sin(rad));
    printf("sec = %.2f\n", 1/cos(rad));
    printf("cot = %.2f\n", 1/tan(rad));
    */


    /*
    // Two numbers are stored in: C, D  // Task:Interchange (swap) their values
    int C, D, temp;

    printf("Enter values of C and D: ");
    scanf("%d %d", &C, &D);

    temp = C;
    C = D;
    D = temp;

    printf("After swapping:\n");
    printf("C = %d, D = %d\n", C, D);
    */


    /*
    // Input: Cost Price (CP) Selling Price (SP),  Find: Profit OR Loss And the amount
    float cp, sp;

    printf("Enter Cost Price: ");
    scanf("%f", &cp);

```

```c
    printf("Enter Selling Price: ");
    scanf("%f", &sp);

    if(sp > cp)
        printf("Profit = %.2f\n", sp - cp);
    else if(cp > sp)
        printf("Loss = %.2f\n", cp - sp);
    else
        printf("No Profit No Loss\n");
    */


    /*
    // nput: any integer , Output: Even or Odd
    int n;

    printf("Enter a number: ");
    scanf("%d", &n);

    if(n % 2 == 0)
        printf("Even number\n");
    else
        printf("Odd number\n");
    */


    /*
    // Input: any year, Output: Leap year or not
    int year;

    printf("Enter a year: ");
    scanf("%d", &year);

    if((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
        printf("Leap Year\n");
    else
        printf("Not a Leap Year\n");
    */



    /*
    // input: 5 digit no and reverse it,  and check reverse no is equal to input no
    int n, original;
    int d1, d2, d3, d4, d5;
    int rev;

    printf("Enter a 5-digit number: ");
    scanf("%d", &n);

```

```c
    original = n;   // store original

    d1 = n % 10;
    n = n / 10;

    d2 = n % 10;
    n = n / 10;

    d3 = n % 10;
    n = n / 10;

    d4 = n % 10;
    n = n / 10;

    d5 = n % 10;

    rev = d1*10000 + d2*1000 + d3*100 + d4*10 + d5;

    printf("Reversed number = %d\n", rev);

    if(original == rev)
        printf("Both are equal (Palindrome)\n");
    else
        printf("Not equal\n");
    */




    /*
    // Input: ages of Ram, Shyam, Ajay;  Output: youngest person
    int ram, shyam, ajay;

    printf("Enter ages of Ram, Shyam and Ajay: ");
    scanf("%d %d %d", &ram, &shyam, &ajay);

    if(ram < shyam && ram < ajay)
        printf("Ram is youngest\n");
    else if(shyam < ram && shyam < ajay)
        printf("Shyam is youngest\n");
    else
        printf("Ajay is youngest\n");
    */




    /*
    // Input: 3 angles of a triangle, Output: Valid triangle or not, (A triangle is valid if: a + b + c = 180)
    int a, b, c;

```

```c
    printf("Enter three angles: ");
    scanf("%d %d %d", &a, &b, &c);

    if(a + b + c == 180)
        printf("Valid Triangle\n");
    else
        printf("Invalid Triangle\n");

    */



    /*
    // Input: any number, Output: absolute value
     int n;

    printf("Enter a number: ");
    scanf("%d", &n);

    if(n < 0)
        n = -n;

    printf("Absolute value = %d\n", n);
    */


    /*
    // Input: any number, Output: absolute value
    int n;
    scanf("%d", &n);
    printf("Absolute value = %d", abs(n)); // abs is build in function for checking absolute value
    */


    /*
    int length, breadth;
    int area, perimeter;

    // Input from user
    printf("Enter length of the rectangle: ");
    scanf("%d", &length);
    printf("Enter breadth of the rectangle: ");
    scanf("%d", &breadth);

    // Calculate area and perimeter
    area = length * breadth;
    perimeter = 2 * (length + breadth);

    // Compare area and perimeter
```

```c
    if(area > perimeter)
        printf("Area (%d) is greater than Perimeter (%d)\n", area, perimeter);
    else if(area == perimeter)
        printf("Area (%d) is equal to Perimeter (%d)\n", area, perimeter);
    else
        printf("Perimeter (%d) is greater than Area (%d)\n", perimeter, area);

    */



    /*
    // Program to check if three points (x1, y1), (x2, y2), and (x3, y3) lie on a straight line (are collinear) using slope comparison.
    int x1, y1, x2, y2, x3, y3;

    // Input three points from user
    printf("Enter x1 y1: ");
    scanf("%d %d", &x1, &y1);

    printf("Enter x2 y2: ");
    scanf("%d %d", &x2, &y2);

    printf("Enter x3 y3: ");
    scanf("%d %d", &x3, &y3);

    // Check collinearity using slope formula: (y2-y1)*(x3-x2) == (y3-y2)*(x2-x1)
    if ((y2 - y1) * (x3 - x2) == (y3 - y2) * (x2 - x1))
        printf("The points are collinear.\n");
    else
        printf("The points are not collinear.\n");

    */



    /*
    // Input: Coordinates of the circle’s center (x, y), radius r, and a point (px, py), Output: Determine if the point lies inside, on, or outside the circle
    float x, y, r;
    float px, py;
    float distance;

    printf("Enter center coordinates (x y) and radius r: ");
    scanf("%f %f %f", &x, &y, &r);

    printf("Enter point coordinates (px py): ");
    scanf("%f %f", &px, &py);

    distance = sqrt(pow(px - x, 2) + pow(py - y, 2));
```

```c

    if(distance < r)
        printf("The point is inside the circle.\n");
    else if(distance == r)
        printf("The point is on the circle.\n");
    else
        printf("The point is outside the circle.\n");
    */


    /*
    // Input: Coordinates of a point (x, y)  Output: Determine if the point lies on: X-axis, Y-axis, Origin
     float x, y;

    printf("Enter point coordinates (x y): ");
    scanf("%f %f", &x, &y);

    if(x == 0 && y == 0)
        printf("The point is at the origin.\n");
    else if(y == 0)
        printf("The point lies on the X-axis.\n");
    else if(x == 0)
        printf("The point lies on the Y-axis.\n");
    else
        printf("The point lies neither on X-axis nor Y-axis.\n");

    */




    /*
    // Input: Coordinates of a point (x, y)  Output: Determine if the point lies on: X-axis, Y-axis, Origin
    float x, y;

    printf("Enter point coordinates (x y): ");
    scanf("%f %f", &x, &y);

    if(x == 0 && y == 0) {
        printf("The point is at the origin.\n");
    } else if(y == 0) {
        printf("The point lies on the X-axis.\n");
    } else if(x == 0) {
        printf("The point lies on the Y-axis.\n");
    } else {
        printf("The point lies neither on X-axis nor Y-axis.\n");
    }

    */
```

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    /*
    // Input: Any year (e.g., 2026), Output: Day of the week on 1st January of that year
    int year, totalDays = 0, leapYears = 0, day;

    printf("Enter a year: ");
    scanf("%d", &year);

    // Count total days from year 1 to (year-1)
    for(int y = 1; y < year; y++) {
        if((y % 4 == 0 && y % 100 != 0) || (y % 400 == 0)) {
            totalDays += 366;
        } else {
            totalDays += 365;
        }
    }

    // Find day of week (01/01/0001 was Monday → index 0)
    day = totalDays % 7;

    // Print day
    if(day == 0) {
        printf("Monday\n");
    } else if(day == 1) {
        printf("Tuesday\n");
    } else if(day == 2) {
        printf("Wednesday\n");
    } else if(day == 3) {
        printf("Thursday\n");
    } else if(day == 4) {
        printf("Friday\n");
    } else if(day == 5) {
        printf("Saturday\n");
    } else {
        printf("Sunday\n");
    }
    */



    /*
    //  Write a program to check if a given number is Armstrong number or not.
    int n, original, remainder, digits = 0;
    double sum = 0;

    printf("Enter a number: ");
    scanf("%d", &n);

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    original = n;

    // Count digits
    while(n != 0) {
        digits++;
        n /= 10;
    }

    n = original;

    // Calculate sum of digits raised to power of digits
    while(n != 0) {
        remainder = n % 10;
        sum += pow(remainder, digits);
        n /= 10;
    }

    // Check and print result
    if(sum == original) {
        printf("%d is an Armstrong number.\n", original);
    } else {
        printf("%d is not an Armstrong number.\n", original);
    }
    */



    /*
    // Write a program to check if the given number is a natural number.
    int n;

    printf("Enter a number: ");
    scanf("%d", &n);

    if(n > 0) {
        printf("%d is a natural number.\n", n);
    } else {
        printf("%d is not a natural number.\n", n);
    }
    */



    /*
    // Degree to Radian Conversion
    #define M_PI  3.141592653589793
    double angle_deg, angle_rad;
    double sine_val, cosine_val, sum_squares;

    // Input angle in degrees
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    printf("Enter angle in degrees: ");
    scanf("%lf", &angle_deg);

    // Convert degrees to radians
    angle_rad = angle_deg * M_PI / 180.0;

    // Calculate sine and cosine
    sine_val = sin(angle_rad);
    cosine_val = cos(angle_rad);

    // Calculate sum of squares
    sum_squares = sine_val * sine_val + cosine_val * cosine_val;

    // Display values
    printf("sin^2(angle) + cos^2(angle) = %.6lf\n", sum_squares);

    // Check if sum of squares is equal to 1 (allowing tiny floating-point error)
    if (fabs(sum_squares - 1.0) < 1e-6) {
        printf("The sum of squares is approximately equal to 1 ✅\n");
    } else {
        printf("The sum of squares is NOT equal to 1 ❌\n");
    }
    */



    /*
    float weight, height, bmi;

    // 1. Input weight and height
    printf("Enter weight in kilograms: ");
    scanf("%f", &weight);
    printf("Enter height in meters: ");
    scanf("%f", &height);

    // 2. Calculate BMI
    // Formula: weight / (height * height)
    bmi = weight / (height * height);

    printf("\nYour BMI is: %.2f\n", bmi);
    printf("Category: ");

    // 3. Determine Category based on the table provided
    if (bmi < 15) {
        printf("Starvation\n");
    }
    else if (bmi >= 15.1 && bmi <= 17.5) {
        printf("Anorexic\n");
    }
    else if (bmi >= 17.6 && bmi <= 18.5) {
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        printf("Underweight\n");
    }
    else if (bmi >= 18.6 && bmi <= 24.9) {
        printf("Ideal\n");
    }
    else if (bmi >= 25 && bmi <= 25.9) {
        printf("Overweight\n");
    }
    else if (bmi >= 30 && bmi <= 30.9) {
        printf("Obese\n");
    }
    else if (bmi >= 40) {
        printf("Morbidly Obese\n");
    }
    else {
        // Handling gaps in the table (e.g., 26 to 29.9)
        printf("Result falls between defined categories.\n");
    }
    */


    /*
    float a, b, c;

    // Input the three sides of the triangle
    printf("Enter the lengths of three sides: ");
    scanf("%f %f %f", &a, &b, &c);

    // 1. Check for Equilateral (All sides equal)
    if (a == b && b == c) {
        printf("The triangle is Equilateral.\n");
    }
    // 2. Check for Isosceles (Any two sides equal)
    else if (a == b || b == c || a == c) {
        printf("The triangle is Isosceles.\n");
    }
    // 3. Check for Scalene (No sides equal)
    else {
        printf("The triangle is Scalene.\n");
    }

    // 4. Separate check for Right-Angled Triangle (Pythagorean theorem: a^2 + b^2 = c^2)
    // We check all combinations in case the longest side isn't entered last.
    if ((a*a + b*b == c*c) || (a*a + c*c == b*b) || (b*b + c*c == a*a)) {
        printf("It is also a Right-angled triangle.\n");
    }
    */


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```c
    /*
    // Calculate overtime pay of 10 employees. Overtime is Rs. 120 per hour for hours worked above 40. Assume no fractional hours.
     int hours, i;
    float overtimePay;
    const int OVERTIME_RATE = 120;
    const int STANDARD_HOURS = 40;

    for(i = 1; i <= 10; i++) {
        printf("Enter hours worked by employee %d: ", i);
        scanf("%d", &hours);

        if(hours > STANDARD_HOURS)
            overtimePay = (hours - STANDARD_HOURS) * OVERTIME_RATE;
        else
            overtimePay = 0;

        printf("Overtime pay for employee %d is: Rs. %.2f\n\n", i, overtimePay);
    }
    */


    /*
    // Find the factorial of any number entered through the keyboard.  Factorial of n (n!) = 1 × 2 × 3 × ... × n   Example: 5! = 1 × 2 × 3 × 4 × 5 = 120
     int num, i, fact;

    printf("Enter a number: ");
    scanf("%d", &num);

    fact = i = 1;

    while(i <= num) {
        fact = fact * i;
        i++;
    }
    printf("Factorial value of %d = %d\n", num, fact);

    */




    /*
    // Compute the value of one number raised to the power of another. Input: two numbers (x as base, y as exponent)  Output: x^y
     float x, power;
    int y, i;

    printf("Enter two numbers (base and exponent): ");
    scanf("%f %d", &x, &y);

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    power = i = 1;

    while(i <= y) {
        power = power * x;
        i++;
    }

    printf("%f to the power %d is %f\n", x, y, power);
    */



    /*
    // Print all ASCII values (0–255) and their equivalent characters using a while loop.
     int i = 0;  // Initialize counter to 0 (ASCII start)

    // Loop from 0 to 255
    while(i <= 255) {
        printf("ASCII %d = %c\n", i, i); // Print value and character
        i++; // Move to next ASCII value
    }
    */



    /*
    // Print all Armstrong numbers between 1 and 500.  Armstrong number: Sum of cubes of each digit = the number itself.  Example: 153 → 1³ + 5³ + 3³ = 153
     int num, original, digit, sum;

    num = 1;

    while(num <= 500) {
        original = num;  // Store the number to compare later
        sum = 0;

        // Sum cubes of digits
        while(original != 0) {
            digit = original % 10;      // Extract last digit
            sum += digit * digit * digit; // Add cube of digit
            original /= 10;             // Remove last digit
        }

        if(sum == num) {
            printf("%d is an Armstrong number\n", num);
        }

        num++; // Move to next number
    }
    */
```

```c



    /*
    // Play a matchstick game between a user and computer.  Rules:  21 matchsticks at start. Each player can pick 1, 2, 3, or 4 sticks.  Player forced to pick the last stick loses. Computer always plays optimally to win.
      int total = 21;  // Total matchsticks
    int userPick, compPick;

    printf("Welcome to Matchstick Game! You lose if you pick last matchstick.\n");

    while(total > 1) {  // Game continues until 1 stick left
        // User turn
        do {
            printf("\nThere are %d matchsticks. Pick 1-4: ", total);
            scanf("%d", &userPick);
        } while(userPick < 1 || userPick > 4 || userPick > total);

        total -= userPick;

        if(total == 1) {  // Last stick left → user loses
            printf("\nYou are forced to pick the last matchstick. You lose!\n");
            break;
        }

        // Computer turn (optimal)
        compPick = 5 - userPick;  // Always leave multiple of 5
        if(compPick > total - 1) {  // Adjust if sticks remaining are less
            compPick = total - 1;
        }

        printf("Computer picks %d matchstick(s)\n", compPick);
        total -= compPick;
    }
    */



    /*
    // Enter numbers repeatedly until the user decides to stop.  Count how many numbers are positive, negative, and zeros.
    int num;
    int positive = 0, negative = 0, zeros = 0;
    char choice;

    do {
        // Take input
        printf("Enter a number: ");
        scanf("%d", &num);

```

```c
        // Count based on value
        if(num > 0)
            positive++;
        else if(num < 0)
            negative++;
        else
            zeros++;

        // Ask user to continue
        printf("Do you want to enter another number? (y/n): ");
        scanf(" %c", &choice); // space before %c to consume newline

    } while(choice == 'y' || choice == 'Y');

    // Display final counts
    printf("\nPositive numbers: %d\n", positive);
    printf("Negative numbers: %d\n", negative);
    printf("Zeros: %d\n", zeros);
    */



    /*
    // Convert a decimal integer to its octal equivalent.  Octal: Base-8 number system (digits 0–7).  Logic: Divide number by 8 repeatedly, store remainders, then reverse them.
    int num, original, remainder;
    int octal[20], i = 0; // array to store remainders

    printf("Enter an integer: ");
    scanf("%d", &num);

    original = num;

    // Convert decimal to octal
    while(num != 0) {
        remainder = num % 8;       // get remainder
        octal[i] = remainder;      // store in array
        i++;
        num = num / 8;             // reduce number
    }

    // Print octal in reverse order
    printf("Octal equivalent of %d = ", original);
    for(i = i - 1; i >= 0; i--) {
        printf("%d", octal[i]);
    }
    printf("\n");
    */


```

```c

    /*
    // Find the range of a set of numbers entered by the user.  Range = Biggest number – Smallest number
    int n, i, num;
    int max, min;

    printf("How many numbers do you want to enter? ");
    scanf("%d", &n);

    // Take first number as initial max and min
    printf("Enter number 1: ");
    scanf("%d", &num);
    max = min = num;

    // Loop through remaining numbers
    for(i = 2; i <= n; i++) {
        printf("Enter number %d: ", i);
        scanf("%d", &num);

        if(num > max)
            max = num;  // Update max
        if(num < min)
            min = num;  // Update min
    }

    printf("Range of the numbers = %d\n", max - min);
    */




    /*
    // Print a rectangle of stars
     int i, j;
    int rows = 3;   // Number of rows
    int cols = 5;   // Number of stars per row

    for(i = 1; i <= rows; i++) {      // Loop for rows
        for(j = 1; j <= cols; j++) {  // Loop for columns
            printf("*");
        }
        printf("\n"); // Move to next line after each row
    }
    */




    /*
    // Check if a given number is prime.  Prime number = greater than 1 and divisible only by 1 and itself.
```

```c
    int num, i, isPrime = 1; // isPrime = 1 means assume prime

    printf("Enter a number: ");
    scanf("%d", &num);

    if(num <= 1) {
        isPrime = 0; // Numbers <= 1 are not prime
    } else {
        for(i = 2; i <= num/2; i++) { // Check divisibility
            if(num % i == 0) {
                isPrime = 0; // Found a divisor → not prime
                break;
            }
        }
    }

    if(isPrime)
        printf("%d is a prime number\n", num);
    else
        printf("%d is not a prime number\n", num);

    */



    /*
    // Print all prime numbers between a given range (start to end).
    int start, end, i, j, isPrime;

    printf("Enter starting number: ");
    scanf("%d", &start);
    printf("Enter ending number: ");
    scanf("%d", &end);

    printf("Prime numbers between %d and %d are:\n", start, end);

    // Loop through the range
    for(i = start; i <= end; i++) {
        if(i <= 1)
            continue; // Skip 0 and 1

        isPrime = 1; // Assume i is prime

        // Check divisibility
        for(j = 2; j <= i/2; j++) {
            if(i % j == 0) {
                isPrime = 0; // Found a divisor → not prime
                break;
            }
        }
```

```c

        if(isPrime)
            printf("%d ", i);
    }

    printf("\n");
    */


    /*
    //  Print all prime numbers from 1 to 300
    int n, i;
    for (n = 1; n <= 300; n++) {
        for (i = 2; i < n; i++) {
            if (n % i == 0)
                break;
        }
        if (i == n && n != 1)
            printf("%d\t", n);
    }
    */



    /*
    //  Sum of first seven terms of the series 1/1! + 2/2! + 3/3! + ...
    int i, j;
    float fact, sum = 0.0;

    for (i = 1; i <= 7; i++) {
        fact = 1.0;
        for (j = 1; j <= i; j++)
            fact *= j;
        sum += i / fact;
    }

    printf("Sum of series = %f\n", sum);
    */


    /*
    // Generate all combinations of 1, 2, 3
    int i, j, k;
    for (i = 1; i <= 3; i++) {
        for (j = 1; j <= 3; j++) {
            for (k = 1; k <= 3; k++) {
                printf("%d %d %d\n", i, j, k);
            }
        }
    }
```

```c
    */




    /*
    int num, square;
    char another;

    //  Execution of a loop an unknown number of times (do-while)
    do {
        printf("Enter a number: ");
        scanf("%d", &num);
        square = num * num;
        printf("Square of %d is %d\n", num, square);

        printf("Want to enter another number y/n: ");
        fflush(stdin);          // Clear input buffer
        scanf(" %c", &another); // Read next character
    } while (another == 'y');
    */



    /*
    int num, i;

    // Program: Print multiplication table of a given number
    printf("Enter a number: ");
    scanf("%d", &num);

    for (i = 1; i <= 10; i++)
        printf("%d * %d = %d\n", num, i, num * i);
    */



    /*
// Program to generate all Pythagorean Triplets with side length <= 30
int main() {
    int a, b, c;
    for (a = 1; a <= 30; a++) {
        for (b = a; b <= 30; b++) {
            for (c = b; c <= 30; c++) {
                if (a * a + b * b == c * c) {
                    printf("(%d, %d, %d)\n", a, b, c);
                }
            }
        }
    }
```

```c
*/



/*
    // Program to determine the town population at the end of each year for the last decade
int main() {
    double pop = 100000;
    printf("Year 10 (Current): %.0f\n", pop);

    for (int i = 9; i >= 1; i--) {
        pop /= 1.1; // Reverse the 10% increase
        printf("Year %d: %.0f\n", i, pop);
    }
*/




/*
// Program to print Ramanujan numbers (sum of two cubes in two different ways)

int main() {
    int limit = 30; // Search limit for base numbers
    for (int n = 1; n <= 50000; n++) {
        int count = 0;
        for (int i = 1; i * i * i < n; i++) {
            for (int j = i + 1; i * i * i + j * j * j <= n; j++) {
                if (i * i * i + j * j * j == n) {
                    count++;
                }
            }
        }
        if (count == 2) {
            printf("%d is a Ramanujan Number\n", n);
        }
    }
        */




        /*
    // Program to print 24 hours of the day with suitable suffixes
int main() {
    for (int i = 0; i < 24; i++) {
        if (i == 0) printf("12 Midnight\n");
        else if (i < 12) printf("%d AM\n", i);
        else if (i == 12) printf("12 Noon\n");
        else printf("%d PM\n", i - 12);
```

```c
    }

    */



    /*
    // Program to produce a centered number pyramid output
int main() {
    int rows = 4, num = 1;
    for (int i = 1; i <= rows; i++) {
        // Print spaces for alignment
        for (int space = 1; space <= rows - i; space++) {
            printf("  ");
        }
        // Print numbers
        for (int j = 1; j <= i; j++) {
            printf("%d   ", num++);
        }
        printf("\n");
    }
*/



    return 0;
}
```